

Technical Summary

Ottermap Open Vision / ML Engineer Intern — 72-Hour Technical Challenge

1. Approach

Model architecture

U-Net with a ResNet34 encoder pretrained on ImageNet (**segmentation_models_pytorch**), fine-tuned for 7-class semantic segmentation. Transfer learning was treated as essential, not optional, given the very small amount of available training data.

Training methodology

Training proceeded through three documented iterations, each motivated by evidence from the previous run rather than blind hyperparameter search:

- **Run 1 (baseline, 40 epochs):** validation mIoU peaked at epoch 3 (0.084) and degraded for the remaining 37 epochs while training loss kept falling — a clear overfitting signature on only 264 training tiles.
- **Run 2 (encoder frozen for epochs 1–10, weight decay, early stopping):** best mIoU 0.062 at epoch 13. A similar ceiling was reached despite the regularization, indicating the limiting factor was data volume, not training strategy.
- **Run 3 / final (weighted oversampling of tiles containing rare classes — road, water, sidewalk, parking_lot):** best mIoU 0.042 — lower in aggregate, but per-class IoU for every rare class improved by 3×–100× (e.g. parking_lot 0.00002 → 0.0025, sidewalk 0.0005 → 0.012), at the cost of turf_grass collapsing (0.026 → 0.00002). This is a genuine bias-variance tradeoff: with this little data, the model cannot learn all 7 classes well simultaneously. This checkpoint was kept deliberately for broader class coverage, since a model that never predicts road/water/parking_lot at all is less useful for a GIS workflow than one with weak but present detections across all classes.

Key technical insight

Per-class performance correlated more strongly with **instance diversity and shape compactness** than with raw pixel count. The *road* class had more total training pixels (953,900) than *woody_vegetation* (265,803) but performed far worse (IoU 0.002 vs 0.18), because road pixels come from very few, thin, elongated real-world segments rather than many varied examples. This suggests pixel-count-based loss weighting alone is insufficient for this kind of imbalance — sample-level oversampling helped more directly, confirmed empirically by comparing Run 2 vs Run 3.

2. Dataset Preparation

Preprocessing

Two non-trivial data-quality issues were discovered and resolved before training was possible, both diagnosed empirically rather than assumed:

- **Missing georeferencing:** the provided GeoTIFFs had no embedded CRS or geotransform (identity matrix), while label polygons were in valid EPSG:4326 coordinates. Reconstructed each image's geotransform from its corresponding label bounding box and verified the fix visually — polygons align tightly with real building/feature outlines across all 3 parcels post-fix.
- **Missing class labels:** all provided polygons (GeoJSON and the similarly-named "Shapefile" export) had only null id/Area attributes — no class information at all, despite the brief stating labels were provided. Used CLIP zero-shot classification to assign each of the 161 polygons to one of 8 candidate classes based on its image crop. Visual and statistical spot-checking revealed CLIP could not reliably distinguish "shrubs" from "tree_canopy" (one acted as a catch-all for the other); these were merged into a single woody_vegetation class, reducing the schema to 7 classes.

Tiling and class distribution

Polygons were rasterized into per-pixel masks and tiled into 256×512 patches per the brief's specification, discarding tiles >95% background (305 of 427 candidate tiles retained). Annotation coverage was also found to be **partial** — only ~48% of parcel 3's labelable area was actually annotated in the source data — meaning real, visually identifiable features (rooftops, lots) outside the original polygons are rasterized as background. This is a genuine source-data limitation, not a pipeline bug, and contributes additional label noise beyond what the small sample size alone would cause.

Train/validation split

Split **by parcel** rather than randomly across tiles: parcel 1 (41 tiles) held out entirely as validation, training on parcels 2+3 (264 tiles). A random tile-level split would leak spatially-correlated information between train and validation (adjacent tiles share

lighting, rooftop style, etc.), inflating the validation score in a way that doesn't reflect genuine generalization to new geography — the primary criterion this challenge is graded on.

Augmentation

Horizontal/vertical flips, brightness/contrast jitter, hue/saturation shift, and light Gaussian noise (Albumentations). 90° rotation was deliberately excluded: it swaps height/width on the non-square 256×512 tiles, breaking batch collation when different samples in a batch ended up with inconsistent shapes — confirmed and fixed during development.

3. Results

Class	Train pixels	Val IoU (best epoch)
background	12,131,977	0.112
building	879,275	0.025
turf_grass	19,938,420	0.00002
woody_vegetation	265,803	0.183
parking_lot	34,315	0.0025
road	953,900	0.0017
sidewalk	306,199	0.012
water	93,119	0.0011

Final validation mIoU: **0.042** (epoch 15/40, early-stopped). For context, this is low in absolute terms — a consequence of training on only 3 source images, not a sign of an implementation defect. Background and woody_vegetation (the two highest-pixel-count classes) perform best; the four rarest classes remain near-zero despite oversampling.

Generalization test (external imagery)

Per the brief's requirement, the model was additionally tested on 3 real, public-domain aerial images sourced directly from the USGS NAIP ImageServer (Austin TX, Denver CO, Miami FL — all dense urban downtown cores, distinct from the suburban/sports-complex training distribution). Qualitative findings: the model substantially over-predicts "water" on shadowed areas between tall buildings (plausibly confusing dark shadow with water, given water was a rare and poorly-learned class), over-predicts "parking_lot" on flat rooftops, and rarely predicts "building" at all despite buildings dominating the scene. This is consistent with and explained by the training diagnosis above: the model was never exposed to multi-story urban buildings, and had weak signal for water and building generally. A known artifact (visible grid-line seams at 256×512 tile boundaries, from independent per-tile inference with no overlap blending) is also present in all outputs.

GIS output quality

Raw per-pixel predictions are noisy; vectorizing them directly produced 23,000+ meaningless micro-polygons per image. Applying a 5×5 median filter before vectorization, plus dropping polygons under 20 square pixels, reduced this to 1,500–2,100 usable features per external image while preserving the predicted regions' overall shape.

4. Improvements With More Time or Data

- More source imagery, especially for road/water/parking_lot/sidewalk — the dominant limitation is the number of distinct real-world instances per class, not model capacity or training strategy.
- Overlapping-tile inference with prediction blending at tile boundaries, to remove the visible grid-seam artifact.
- A foundation-model zero-shot baseline (SAM for region proposals + CLIP for classification) as a direct comparison point against the trained supervised model, particularly relevant given how little labeled data was actually available.
- Manual correction of the lowest-confidence CLIP pseudo-labels, rather than relying on automatic labeling alone, to reduce training label noise.
- Dense/full annotation of the source parcels (current coverage is only ~48% in at least one parcel), which would remove a meaningful source of background-class label noise independent of the small-sample-size issue.